Problem Statement

Customer Churn Prediction Using Machine Learning

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Customer churn is one of the biggest challenges faced by subscription-based businesses. Acquiring a new customer is significantly more expensive than retaining an existing one. Therefore, companies need a reliable way to identify customers who are likely to discontinue their services so that proactive retention strategies can be implemented.

The objective of this project is to develop a machine learning model that predicts whether a customer is likely to churn based on customer demographics, subscription details, and spending behavior.

The project involves:

- Exploring and understanding customer data.
- Performing data preprocessing and feature engineering.
- Visualizing important patterns and correlations.
- Training multiple machine learning classification models.
- Comparing model performance using evaluation metrics.
- Selecting the best-performing model for churn prediction.
- Interpreting the model through feature importance analysis.

The final solution helps businesses identify high-risk customers, improve customer retention strategies, reduce revenue loss, and make data-driven business decisions.